Engineering Mathematics - III [20MA312]

Unit-I: Statistics

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- Introduction and Definitions:

* Statistics deals with collecting, analyzing and classification of data.
* Statistics is the science of facts and figures that may be readily available or obtained through direct enquiry or enumeration. It deals with method of collecting, classifying and analyzing the data so as to draw valid conclusions.

* Classification:

The significance of a large mass of statistical data known as raw data, which cannot be understood, unless it is arranged in some definite manner. the process of arrangement in different groups are called classification.

* Frequency Table:

It is a tabular arrangement consisting of various classes of uniform size known as class intervals and the number in each class is known as frequency. The difference two consecutive vertical enteries of classes is known as the width of class, usually denoted by 'h'. The average of left and right endpoints of the class intervals is known as the mid-point of class, denoted by 'x'

* **Variate:**

It is a quantity which can have different numerical value.

• **Basic formulae:**

* **Mean ($\bar{x}$):**

If $x_1, x_2, \cdots, x_n$ are the n set of values, then the mean of given data is given by

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

For grouped data, $\bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$

* **Variance (V):**

For the same set of data, variance is given by

$$V = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}$$

For grouped data, $V = \frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}$

* **Standard deviation (σ):**

$$\sigma = \sqrt{V}$$

1.) Find the mean and standard deviation for following grouped data.

Class	1-10	11-20	21-30	31-40	41-50	51-60
frequency	3	16	26	31	16	8

Ans:

Class	frequency (f_i)	x_i	$f_i x_i$	$f_i(x_i-\bar{x})^2$
1-10	3	5.5	16.5	816.75
11-20	16	15.5	248.0	
21-30	26	25.5	663.0	
31-40	31	35.5	1100.5	
41-50	16	45.5	728.0	
51-60	8	55.5	444.0	

$\Sigma f_i = 100$ $\Sigma x_i =$ $\Sigma f_i x_i = 3199$ $\Sigma f_i(x_i-\bar{x})^2 = 15275$

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i(x_i)}{\sum_{i=1}^{n} f_i} = \frac{3190}{100} = 31.99 \approx 32$$

$$V = \frac{\Sigma f_i(x_i-\bar{x})^2}{\Sigma f_i} = \frac{15275}{100} = 152.75$$

$$\sigma = \sqrt{V} = \sqrt{152.75} = 12.36$$

• Curve Fitting:

The method of finding a specific equation $y = f(x)$ for the data to satisfy as accurately as possible and such an equation is called best fitting equation or the curve of best fit.

The method is called as the method of least square.

→ Least square method:

Suppose $y = f(x)$ is an approximate equation that fits in the given data, x_i, y_i where $i = 1, 2, 3, \ldots, n$. Here y_i are the observed values and Y_i are called expected value. Their difference $R_i = y_i - Y_i$ are called the residuals or estimate errors.

The method of least squares provides a relationship $y = f(x)$ such as the sum of square of residuals is least or minimum.

→ Fitting of a straight line $y = a + bx$:

* Consider a set of n given values, (x, y) for fitting the straight line $y = a + bx$, where a, b are parameters that are to be determined.

The residual, $R = y - (a + bx)$.

It is the difference b/w observed and expected values of y. By the method of least square, we can find parameters a, b, such that the sum of square of residuals are least or minimum.

Let $S = \sum_{i=1}^{n} R_i^2$

$\Rightarrow S = \sum_{i=1}^{n} (y-(a+bx))^2$ ——①

Treating S as a function of two parameters a and b, the necessary conditions for S to be minimum is,

$$\frac{\partial S}{\partial a} = 0 \; ; \; \frac{\partial S}{\partial b} = 0$$

Differentiating eqn.① partially w.r.t. a, and b,

$$\frac{\partial S}{\partial a} = \sum_{1}^{n} 2(y-(a+bx))(-1)) = 0$$

$$\frac{\partial S}{\partial b} = \sum_{1}^{n} 2(y-(a+bx))(-x) = 0$$

Diving both equation by 2 and rearranging,

$$\boxed{\begin{aligned} na + b\sum_{1}^{n} x &= \Sigma y \\ a\Sigma x + b\Sigma x^2 &= \Sigma yx \end{aligned}}$$

x values - arguments

→ Step deviation method:

This is appropriate when the values of x are larger in magnitude and are equally spaced.

↳ sign is not considered.

* To fit a curve $y=f(x)$, in the form of a straight line or parabola, we choose $Y = y - \bar{y}$, where $\bar{y}$ is rounded off to the nearest integer.
* we choose $X = \frac{x-a}{h}$, when n is odd, where a is middle value of x.

or $X = \frac{x - \frac{1}{2}(a+b)}{h/2}$, when n is even, where a & b are the two middle values of x and h is common difference in values of x

(6)

* $\Sigma x = 0$. $\Sigma x^3 = 0$.

→ Find the equation of the best fitting straight line $y = ax + b$ for the following data.

x	y
5	16
10	19
15	23
20	26
25	30

Ans:

Method I: Least square method:

Note: Normal equations $\Sigma y = nb + a\Sigma x$ } for $y = ax + b$
$\Sigma yx = b\Sigma x + a\Sigma x^2$ } fitting

Here: $n = 5$

x	y	yx	x^2
5	16	80	25
10	19	190	100
15	23	345	225
20	26	520	400
25	30	750	625

$\Sigma x = 75$ $\Sigma y = 114$ $\Sigma yx = 1885$ $\Sigma x^2 = 1375$

From normal equations,

$\Rightarrow 114 = 5b + 75a$

$1885 = 75b + 1375a$

$b = 12.3$, $a = 0.7$

$\therefore y = 0.7x + 12.3$

Method II: Step deviation method:

x	y
5	16
10	19
15	23
20	26
25	30

Since n is odd, $X = \frac{x-a}{h}$, here $h = 5$ $(n=5)$

As middle value of x is 15, $X = \frac{x-15}{5}$

and $Y = y - \bar{y}$

$\bar{y} = \frac{\Sigma y}{n} = \frac{114}{5} = 22.8 \approx 23$

The normal equations are $\Sigma Y = a\Sigma X + nb$
$\Sigma YX = b\Sigma X + a\Sigma X^2$ } $Y = aX + b$

x	y	X	Y	YX	X^2
5	16	-2	-7	14	4
10	19	-1	-4	4	1
15	23	0	0	0	0
20	26	1	3	3	1
25	30	2	7	14	4
$\Sigma x = 75$	$\Sigma y = 114$	$\Sigma X = 0$	$\Sigma Y = -1$	$\Sigma YX = 35$	$\Sigma X^2 = 10$

$-1 = a(0) + 5b$

$35 = b(0) + a(10)$

$a = 3.5$, $b = -0.2$

$Y = 3.5X + (-0.2)$

$Y = 3.5X - 0.2$

$(y - \bar{y}) = 3.5\left(\frac{x-15}{5}\right) - 0.2$

$\Rightarrow y - 23 = 0.7x - 10.5 - 0.2$

$\Rightarrow y = 0.7x + 12.3$

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2.) Fit a straight line $y = a + bx$ in least square sense for data

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

Ans:

x	y	xy	x^2
1	1	1	1
3	2	6	9
4	4	16	16
6	4	24	36
8	5	40	64
9	7	63	81
11	8	88	121
14	9	126	196

$\Sigma x = 56 \quad \Sigma y = 40 \quad \Sigma xy = 364 \quad \Sigma x^2 = 524$

Normal equations are:

$$\Sigma y = na + b\Sigma x$$

$$\Sigma yx = a\Sigma x + b\Sigma x^2$$

Substituting in normal equation,

$$40 = 8a + 56b$$

$$364 = 56a + 524b$$

$\Rightarrow a = 0.5454 \quad b = 0.6363$

$\therefore y = 0.5454 + 0.6363x$

→ Fitting of a curve $y = ae^{bx}$ [exponential curve]:

Let $y = ae^{bx}$

Taking $\log_e$ on both sides,

$$\log y = \log a + bx \log_e e$$

Since $\log_e e = 1$,

$$\log y = \log a + bx \quad \text{—①}$$

Let $Y = \log_e y$ and $A = \log_e a$.

then, ① $\Rightarrow Y = A + bx$

This is a straight line eqn.

∴ The normal equations are:

$$\Sigma Y = nA + b\Sigma x$$

$$\Sigma Yx = A\Sigma x + b\Sigma x^2$$

1) Fit a curve of form $y = ae^{bx}$ for following.

x	0	2	4
y	8.12	10	31.82

Ans! The normal equations, considering $Y = \log_e y$ and $A = \log_e a$

are $\Sigma Y = nA + b\Sigma x$

$$\Sigma Yx = A\Sigma x + b\Sigma x^2$$

x	y	Y	Yx	x^2
0	8.12	2.094	0	0
2	10	2.302	4.604	4
4	31.82	3.460	13.840	16.

$\Sigma x = 6$ $\Sigma y = \Sigma Y = 7.856$ $\Sigma x^2 = 20.$

$\Sigma Yx = 18.444$

⇒ The normal eqns are:

$$7.856 = 3A + 6b$$

$$18.444 = 6A + 20b$$

$$A = 1.9356$$

$$b = 0.3415$$

$$a = e^A \Rightarrow a = 6.9281$$

∴ Best fitting curve is $y = 6.9281\, e^{0.3415x}$.

1 (10)

∴ Best curve is

$$y = 6.93 e^{0.3415x}$$

2.) Find the equation of best fitting curve $y = ae^{bx}$ for the data

x	5	6	7	8	9	10
y	133	55	23	7	2	2

Ans: The normal equations considering $Y = \log_e y$ and $A = \log_e a$, are

$$\Sigma Y = nA + b\Sigma x$$

$$\Sigma Yx = A\Sigma x + b\Sigma x^2$$

x	y	Y	Yx	x^2
5	133	4.89	24.45	25
6	55	4.007	24.042	36
7	23	3.135	21.945	49
8	7	1.945	15.56	64
9	2	0.693	6.237	81
10	2	0.693	6.930	100

$\Sigma x = 45$ $\Sigma y = 222$ $\Sigma Yx = 99.164$ $\Sigma x^2 = 355$

$\Sigma Y = 15.363$

By Normal equations,

$$15.363 = 6A + 45b$$

$$99.164 = 45A + 355b.$$

$$\Rightarrow A = 9.442$$

$$b = -0.917.$$

$$A = \log_e a \Rightarrow a = e^A \Rightarrow a = e^{9.442} = 12606 \times 10^{4}$$

∴ The best fitting curve is given by the equation.

$$y = 12,606.9x\, e^{-0.917x}$$

→ Fitting of a second degree parabola of the form $y = a + bx + cx^2$.

Consider a set of n given values (x, y), the residual R for the given equation can be written as

$$R = y - (ax + bx + cx^2)$$

In the method of least square, the sum of squares of residual must be minimum.

$$\Rightarrow S = (y - (ax + bx + cx^2))^2 \quad \text{---} (1).$$

Treating S as a ~~four~~ equation with three parameters a, b, c, the conditions necessary for S to be minimum are:

$$\frac{\partial S}{\partial a} = 0, \quad \frac{\partial S}{\partial b} = 0, \quad \frac{\partial S}{\partial c} = 0.$$

By applying necessary conditions, we get the normal equations:

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$
$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$
$$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

$$\frac{\partial S}{\partial a} = 0 \Rightarrow 2(y - (ax + bx + cx^2)($$

(12)

If the curve is of the form $Y = AX^2 + BX + C$,

$$\Sigma Y = nC + B\Sigma x + A\Sigma x^2$$
$$\Sigma XY = C\Sigma x + B\Sigma x^2 + A\Sigma x^3$$
$$\Sigma X^2Y = A\Sigma x^4 + B\Sigma x^3 + C\Sigma x^2$$

11) Fit a second degree parabola $y = a + bx + cx^2$ for the data

x	0	1	2	3	4
y	1	1·8	1·3	2·5	2·3

Ans:

	x	y	xy	x^2	x^2y	x^3	x^4
	0	1	0	0	0	0	0
	1	1·8	1·8	1	1·8	1	1
	2	1·3	2·6	4	5·2	8	16
	3	2·5	7·5	9	22·5	27	81
	4	2·3	9·2	16	36·8	64	256
Σ	$\Sigma x = 10$	$\Sigma y = 8.9$	$\Sigma 21.1$	30	66·3	100	354

The normal equations for $y = a + bx + cx^2$ are

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$
$$\Sigma yx = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$
$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

$\therefore$ The equations are

$$8.9 = 5a + 10b + 30c \quad \Rightarrow a = 1.077$$
$$21.1 = 10a + 30b + 100c \quad b = 0.415$$
$$66.3 = 30a + 100b + 354c \quad c = -0.0214$$

The best fitting curve is given by

$$y = 1.077 + 0.415x - 0.0214x^2$$

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2) Fit a parabola in the form $y = Ax^2 + Bx + C$ for the following data

x	y
1	10
2	12
3	13
4	16
5	19

Apply step deviation method.

Soln: The total number of obs., $n = 5$, i.e; odd.

$\therefore$ Given $y = Ax^2 + Bx + C$, by Step deviation method,

Let $X = \frac{x-a}{h}$, $Y = y - \bar{y}$ $\qquad \bar{y} = \frac{\Sigma y}{5} = \frac{70}{5} = 14$ //

$\Rightarrow X = \frac{x-3}{1}$, $Y = y - 14$

The normal equations are

$$\Sigma Y = A\Sigma X^2 + B\Sigma X + Cn$$
$$\Sigma XY = A\Sigma X^3 + B\Sigma X^2 + C\Sigma X$$
$$\Sigma X^2 Y = A\Sigma X^4 + B\Sigma X^3 + C\Sigma X^2$$

(14)

x	y	$X = x-3$	$Y = y-\bar{y}$	XY	X^2	X^2Y	X^3	X^4
1	10	-2	-4	8	4	-16	-8	+16
2	12	-1	-2	2	1	-2	-1	1
3	13	0	-1	0	0	0	0	0
4	16	1	2	2	1	2	1	1
5	19	2	5	10	4	20	8	16
15	70	0	0	22	10	4	0	34

∴ The normal eqns are:

$$0 = 10A + B(0) + 5C$$
$$22 = (0)A + B(10) + C(0)$$
$$4 = 34A + B(0) + C(10)$$

$$\Rightarrow \quad 10A + 5C = 0$$
$$34A + 10C = 4$$
$$10B = 22 \quad \Rightarrow \boxed{B = 2.2}$$

$$\Rightarrow \boxed{\begin{aligned} A &= 0.2857 \\ C &= -0.5714 \end{aligned}}$$

∴ The equation is $Y = 0.2857X^2 + 2.2X - 0.5714$

$$\Rightarrow (y-14) = (0.2857)(x-3)^2 + 2.2(x-3) - 0.5714$$

$$\Rightarrow (y-14) = 0.2857x^2 - 1.7142x + 9 + 2.2x - 6.6 - 0.514$$

$$\Rightarrow y - 14 = 0.2857x^2 + 0.4858x + 1.886$$

• Correlation and correlation coefficient:

★ **Correlation**: Co-variance of two independent magnitudes is called as correlation. If two variables x and y are related in such a way that the increase and decrease in anyone of them corresponds to the increase and decrease in the others. then, the variables are +vely correlated.

Then, if increase and decrease of any one, corresponds to decrease and increase, they are −vely correlated.

The numerical measure of correlation b/w x and y is known as Pearson's coefficient of correlation, denoted by r, defined as

$$r = \frac{\sum_1^n (x-\bar{x})(y-\bar{y})}{n\,\sigma_x\,\sigma_y}$$

Let $X = (x-\bar{x})$ and $Y = (y-\bar{y})$

$$\sigma_x^2 = \frac{\Sigma(x-\bar{x})^2}{n}, \quad \sigma_y^2 = \frac{\Sigma(y-\bar{y})^2}{n}$$

$$\Rightarrow \sigma_x^2 = \frac{\Sigma X^2}{n}, \quad \sigma_y^2 = \frac{\Sigma Y^2}{n}$$

Substituting, $$r = \frac{\sum_1^n XY}{\sqrt{\Sigma X^2}\sqrt{\Sigma Y^2}}$$

The correlation coefficient numerically, does not exceed unity, i.e., $-1 < r < 1$

Alternate formula to find coefficient of correlation,

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{xy}^2}{2\sigma_x\sigma_y}.$$

- Regression :

* It is the estimation of one independent variable in terms of another.

* If x and y are correlated, the best fitting straight line in least square sense gives good relation b/w x & y. The best fitting straight line of the form $y = ax + b$, where x being an independent variable, is called regression line y on x.

* Similarly, if st. line is of form $x = ay + b$, where y is the independent variable, is called regression line x on y.

The equation of regression lines:

(i) $y - \bar{y} = r\frac{\sigma_y}{\sigma_x}(x - \bar{x})$, is called regression line y on x

(ii) $(x - \bar{x}) = r\frac{(\sigma_x)}{\sigma_y}(y - \bar{y})$ is called regression line x on y

The coefficients of x and y in (i) and (ii) are known as regression coefficients.

their products is equal to r^2.

Note:

★ Let $r = \pm\sqrt{\text{coeff. of } x \times \text{coeff. of } y}$,

(Since r is the geometric mean of regression coefficients)

★ The lines of regression can also be written in the form

$$Y = \frac{\Sigma XY}{(\Sigma X^2)}(X)$$

$$X = \frac{\Sigma XY}{(\Sigma Y^2)}(Y)$$

★ To compute coefficient of correlation, we use the formula

$$\sigma_x^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2, \quad \sigma_y^2 = \frac{\Sigma y^2}{n} - (\bar{y})^2$$

★ If $\bar{x}$ and $\bar{y}$ are integers, then $r = \frac{\Sigma XY}{\sqrt{\Sigma X^2}\sqrt{\Sigma Y^2}}$

Examples:

1.) Compute coefficient of correlation and eqns. of lines of regression for data

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

Ans: For the given data, coefficient of correlation,

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

Let $z = x - y$.

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$\frac{7(7-1)}{2}$

x	y	z	x^2	y^2	z^2
1	9	-8	1	81	64
2	8	-6	4	64	36
3	10	-7	9	100	49
4	12	-8	16	144	64
5	11	-6	25	121	36
6	13	-7	36	169	49
7	14	-7	49	196	49

$\Sigma x = 28 \quad \Sigma y = 77 \quad \Sigma z = -49$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{28}{7} = 4$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{77}{7} = 11$$

$$\bar{z} = \frac{\Sigma z}{n} = \frac{-49}{7} = -7$$

$\Sigma x^2 = 140$

$\Sigma y^2 = 875$

$\Sigma z^2 = 347$

$$\sigma_x^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2 = \frac{(140)}{7} - (4)^2 = 20 - 16$$

$\Rightarrow \sigma_x^2 = 4 \Rightarrow \sigma_x = 2$

$$\sigma_y^2 = \frac{\Sigma y^2}{n} - (\bar{y})^2 = \frac{875}{7} - (11)^2 = 125 - 121$$

$\sigma_y^2 = 4 \Rightarrow \sigma_y = 2$

$$\sigma_z^2 = \frac{\Sigma z^2}{n} - (\bar{z})^2 = \frac{347}{7} - (-7)^2$$

$= 49.571 - 49 = 0.571$

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_z^2}{2\sigma_x\sigma_y} = \frac{4 + 4 - 0.571}{2(2)(2)} = 0.9285$$

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 11 = (0.9285)\left(\frac{2}{2}\right)(x - 4)$$

$$\Rightarrow y - 11 = 0.9285x - 3.714$$

$$\Rightarrow y = 0.9285x + 7.286$$

2.) Obtain the lines of regression and hence find the coefficient of correlation for the following data:

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

Ans:

x	y
1	9
2	8
3	10
4	12
5	11
6	13
7	14

$\bar{x} = 4$

2.) Obtain the lines of regression and hence find coefficient of correlation.

x	1	3	4	2	5	8	9	10	13	15
y	8	6	10	8	12	16	16	10	32	32

Ans.

x	y	$x - \bar{x}$	Y	x^2	y^2	xY
1	8	-6	-7	36	49	42
3	6	-4	-9	16	81	36
4	10	-3	-5	9	25	15
2	8	-5	-7	25	49	35
5	12	-2	-3	4	9	6
8	16	1	1	1	1	1
9	16	2	1	4	1	2
10	10	3	-5	9	25	-15
13	32	6	17	36	289	102
15	32	8	17	64	289	136

$\Sigma x = 70$ $\Sigma y = 150$ $\Sigma x^2 = 204$ $\Sigma y^2 = 818$ $\Sigma xY = 360$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{70}{10} = 7$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{150}{10} = 15$$

we know that lines of regression are

$$Y = \frac{\Sigma xY}{\Sigma x^2}(x) \quad \text{and} \quad X = \frac{\Sigma xY}{\Sigma Y^2}(Y)$$

$$r = \pm\sqrt{(\text{coeff of } x)(\text{coeff. of } Y)}$$

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$Y = y - \bar{y}$

$X = x - \bar{x}$

$$Y = \frac{\Sigma XY}{\Sigma X^2}(X)$$

$$\Rightarrow (y - \bar{y}) = \frac{\Sigma XY}{\Sigma X^2}(x - \bar{x})$$

$$\Rightarrow (y - 15) = \frac{360}{204}(x - 7)$$

$$y - 15 = 1.764x - 12.352$$

$$\Rightarrow y = 1.764x + 2.648$$

$\therefore$ coefficient of $x = 1.764$

$$X = \frac{\Sigma XY}{\Sigma Y^2}(Y)$$

$$\Rightarrow (x - \bar{x}) = \frac{\Sigma XY}{\Sigma Y^2}(y - \bar{y})$$

$$\Rightarrow (x - 7) = \frac{360}{818}(y - 15)$$

$$\Rightarrow x - 7 = 0.44y - 6.6$$

$$\Rightarrow x = 0.44y + 0.4$$

coeff. of $y = 0.44$

$$r = \pm \sqrt{(\text{coeff. of } x)(\text{coeff. of } y)}$$

$$= \pm \sqrt{(1.764)(0.44)}$$

$$= \pm 0.88$$

3) The following data gives age of husband (x) and age of wife (y) in years. Forms the two regression lines and calculate the age of husband corresponding to 160 years age of wife

Ans:

x	y	X	Y	X^2	Y^2	XY
36	29	6	4	36	16	24
23	18	-7	-7	49	49	49
27	20	-3	-5	9	25	15
28	22	-2	-3	4	9	6
28	27	-2	2	4	4	-4
29	21	-1	-4	1	16	4
30	29	0	4	0	16	0
31	27	1	2	1	4	2
33	29	3	4	9	16	12
35	28	5	3	25	9	15
$\Sigma x = 300$	$\Sigma y = 250$			$\Sigma X^2 = 138$	$\Sigma Y^2 = 164$	123

$$\bar{x} = \frac{\Sigma x}{n} = \frac{300}{10} = 30$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{250}{10} = 25$$

The regression lines are

$$Y = \frac{\Sigma XY}{\Sigma X^2}(X)$$

$$\Rightarrow (y-25) = \frac{123}{138}(x-30)$$

$$\Rightarrow y - 25 = 0.8913x - 26.739$$

$$\Rightarrow y = 0.8913x - 1.739$$

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$$x = \frac{\Sigma xy}{\Sigma y^2}(y)$$

$$\Rightarrow (x-30) = \frac{123}{164}(y-25)$$

$$\Rightarrow x-30 = 0.75y - 18.75$$

$$\Rightarrow x = 0.75y + 11.25 \quad \text{— (2)}$$

For $y = 16$, x is given by subs in (1),

$$x = 0.75(16) + 11.25$$

$$\Rightarrow x = 12 + 11.25$$

$$\Rightarrow x = 23.25$$

$$r = \pm\sqrt{(0.8913)(0.75)}$$

$$= \pm 0.8176$$

4.)

- **Angle between regression lines:**

$$b_{yx} = r\frac{\sigma_y}{\sigma_x}$$

$$b_{xy} = r\frac{\sigma_x}{\sigma_y}$$

We know that the equation of regression lines y on x and x on y as follows:

$$(y-\bar{y}) = r\frac{\sigma_y}{\sigma_x}(x-\bar{x}) \quad \text{---} ①$$

$$(x-\bar{x}) = r\frac{\sigma_x}{\sigma_y}(y-\bar{y}) \quad \text{---} ②$$

here $b_{yx} = r\frac{\sigma_y}{\sigma_x}$ and $b_{xy} = r\frac{\sigma_x}{\sigma_y}$ are called regression coefficients.

$$① \Rightarrow (y-\bar{y}) = \frac{1}{r}\frac{\sigma_y}{\sigma_x}(x-\bar{x}) \quad \text{---} ③$$

Comparing ② & ③ with $y = m_1x + c_1$ and $y = m_2x + c_2$,

$$m_1 = r\frac{\sigma_y}{\sigma_x} \quad \text{and} \quad m_2 = \frac{1}{r}\frac{\sigma_y}{\sigma_x} \quad \text{---} ④$$

If the angle b/w two lines is acute, w.k.t.,

$$\tan\theta = \pm\frac{m_2 - m_1}{1 + m_1m_2}$$

From ④, $\tan\theta = \dfrac{\frac{1}{r}\frac{\sigma_y}{\sigma_x} - r\frac{\sigma_y}{\sigma_x}}{1 + \frac{\sigma_y^2}{\sigma_x^2}}$

$$\Rightarrow \tan\theta = \frac{\frac{\sigma_y}{\sigma_x}\left(\frac{1}{r} - r\right)}{\frac{\sigma_x^2 + \sigma_y^2}{\sigma_x^2}}$$

$$\Rightarrow \tan\theta = \frac{\sigma_y \sigma_x (1-r^2)}{r(\sigma_x^2 + \sigma_y^2)}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{\sigma_y \sigma_x (1-r^2)}{r(\sigma_x^2 + \sigma_y^2)} \right)$$

Note: when the angle b/w two regression lines,

$$\theta = \tan^{-1}\left(\frac{\sigma_y \sigma_x (1-r^2)}{r^2(\sigma_x^2 + \sigma_y^2)} \right),$$

Interpret the result $r = 0$ and $r = 1$ or -1

when $r = 0$, $\tan\theta = \infty$, $\Rightarrow \theta = \pi/2$. So the lines of regression are perpendicular to each other.

when $r = 1$, $\tan\theta = 0$, $\Rightarrow \theta = 0$. Therefore the two lines coincide ($+^{ve}$ correlation)

when $r = -1$, $\tan\theta = 0$, $\Rightarrow \theta = \pi$. Therefore the two lines are negetively correlated.

1.) If tangent of angle θ b/w lines of regression of y on x and x on y is 0.6. And Standard deviation of y is twice the S.D. of x, then find coeff. of correlation b/w x and y.

Ans: **Given**: $\tan\theta = 0.6$

$\sigma_y = 2\sigma_x$

To find: $r = ?$

Solution: W.K.T. $\tan\theta = \frac{\sigma_y \sigma_x (1-r^2)}{r(\sigma_x^2 + \sigma_y^2)}$

$$\Rightarrow 0.6 = \frac{(2\sigma_x)(\sigma_x)(1-r^2)}{r(\sigma_x^2 + 4\sigma_x^2)}$$

$$\Rightarrow 0.6 = \frac{2\sigma_x^2(1-r^2)}{5\sigma_x^2 r}$$

$$\Rightarrow 0.6 = \frac{2}{5}\frac{(1-r^2)}{r}$$

$$\Rightarrow 3r = 2 - 2r^2$$

$$\Rightarrow 2r^2 + 3r - 2 = 0$$

$$\Rightarrow r = \frac{1}{2}, -2.$$

As r cannot be greater than 1 and less than -1,

$$r = 0.5$$